

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

Code:

```
clc;
clear;
close all;

% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);

% Decimation by 2
y = x(1:2:end);
nd = 0:length(y)-1;

% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;

% Figure 1: Original Sequence
figure;
stem(n,x,'filled');
grid on;
title('Original Sequence x[n]);
```

```
xlabel('n');  
ylabel('Amplitude');
```

```
% Figure 2: Decimated Sequence
```

```
figure;  
stem(nd,y,'filled');  
grid on;  
title('Decimated Sequence by 2');  
xlabel('n');  
ylabel('Amplitude');
```

```
% Figure 3: Interpolated Sequence
```

```
figure;  
stem(ni,v,'filled');  
grid on;  
title('Interpolated Sequence by 2');  
xlabel('n');  
ylabel('Amplitude');
```

```
% Figure 4: Comparison
```

```
figure;  
stem(n,x,'b');  
hold on;  
  
stem(0:2:length(v)-2,v(1:2:end),'r');
```

```
legend('Original x[n]', 'Interpolated Samples');
```

```
grid on;
```

```
title('Comparison of Sequences');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

Output:



